

Lecture 7: The exam, and how to prepare for it

Energy Economics of the Green Transition

Rasmus K. S. Berg

Monday 26 October 2026

Department of Economics, University of Copenhagen

Today

1. The exam's format
2. What the questions look like
3. How answers are marked
4. Mock exams, and the Model Lab's exams

The format

The exam: three hours, closed book, on a computer

- **Three hours, at the department.**
- **No aids:** no notes, no internet, no AI, no programming.
- **On a computer:** you type your answers; maths in Word or LaTeX. A diagram can be drawn by hand or in any program.
- **Examinable:** the required readings (the note's sections and the papers on the reading list), the slides, and the exercises with their solutions.

Two or three parts, 100 points

- **One or two model parts**, and **an essay part**.
- **Two model parts:** one on lectures 3–7, one on lectures 8–13, and three essays on lectures the model parts do not use.
- **One model part:** five essays carry the rest of the course.
- **Points are printed with every question** and sum to 100. Parts weigh equally, unless their lengths clearly differ.

Lectures 1 and 2 enter at most as part of a question.

What the questions look like

A model part builds a model up, question by question

- **Maths, interpretation, and evidence or policy**, mixed.
- **No question blocks the rest:** results you are asked to show are stated, so you can carry on without them.
- **The last question is the least guided:** an extension or a new setting.
- **Evidence is printed with the exam:** a paper's figure or table, or a Model Explorer figure.

An essay asks for a mechanism or an argument, in about 200 words

- **A word limit** on every essay, about 200 words; longer where a question needs it.
- **A mechanism, an argument or an assessment**, answered from the course's results.
- **No number and no reference** is needed for a full answer.

Mock exam 1, C.2: *Do heat pumps help integrate wind power? Discuss what a growing fleet of heat pumps does to the cost of the electricity they use and to the market value of wind power, and what making the heat pumps flexible changes.*

How answers are marked

Every question has a solution and a marking guide

- **The points an answer must make**, mechanisms first, each with its share of the question's points.
- **Points that support an answer:** they make it better, but a full answer does not need them.
- **What an answer at each level does**, from full marks to none.
- **A mechanism counts more than a cited result**, and a little more than an example.

The solutions may cite papers and numbers; your answer does not have to.

An example: mock exam 1, A.1 (4 points)

Explain the difference between a direct and an indirect network effect, and why the relation between EVs and charging stations is an indirect one. Why might it be a reason for policy to intervene?

An answer must make	points
An indirect effect runs through a complementary good, a direct one through users' utility	1.5
EVs raise the return to chargers, and chargers raise the value of EVs: a two-sided feedback	1.5
Neither side takes its effect on the other into account, so adoption may be too low	1

Mock exams, and the Model Lab’s exams

Mock exams: the first is out, with its solutions

- **Mock exam 1** and its solutions with the marking guide: *exams/* in the course repository.
- **Three mocks are planned**; solutions are published with each, nothing held back.
- **Every mock says where it comes from:**
  - **AI-generated:** written by an AI model from written instructions on how the exam is structured and what it examines;
  - **Past exam**;
  - **Reviewed:** a mock your teacher has been through.
- **Mock 1 is AI-generated**, from the 2025 exam, adjusted for length and difficulty.

In the Model Lab: the Exams page

- **The same exam as the PDF:** the same questions, order and points. Answer the parts in any order; submit the exam whole.
- **The AI tutor reads each open answer:** feedback, and a number out of 100. Its estimate, not a grade, and not yet calibrated on exam questions.
- **Check a draft** with the tutor before you submit; after, every question shows its model answer and marking guide.

`championape-model-lab.share.connect.posit.cloud`, under *Exams*.